Examiners' Report
Summer 2026

Pearson Edexcel GCE

In AS Further Mathematics (8FM0)

Paper 22 Further Pure Mathematics 2

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General Comments

This paper provided learners with opportunities to demonstrate their understanding across the specification content. Many learners were able to access the familiar styles of question and showed secure knowledge of standard methods.

There were also some less familiar elements that required learners to apply their understanding in more demanding contexts. The final question provided appropriate challenge for higher-attaining learners, with the strongest responses showing clear reasoning and well-structured working.

Individual Question Comments

Question 1

Part (a)

A straightforward opening task, requiring learners to complete a Cayley table, gave most learners the opportunity to make a positive start to the paper. Correct completions of the table were seen in most responses, with only a small number of learners unable to make progress on this topic.

It was, however, common for learners to write the entries in the table in full as complex numbers, rather than using the assigned labels. This often made their working longer than necessary.

The remaining parts of the questions, (b) and (c), produced a slightly more mixed response, although most learners were able to make progress across them.

Part (b)

For part (b), most learners showed a good understanding of the group properties that needed to be satisfied. The most common reason for losing a mark was not identifying the inverses sufficiently clearly.

Some learners gave a standard statement of the inverse property without relating it to the group in the question, which meant the second B mark could not be awarded. The identity element was usually identified correctly, and closure and associativity were generally checked or commented on appropriately before a conclusion was given.

Part (c)

Part (c) was sometimes omitted, possibly through oversight. Where it was attempted, responses were generally correct, with many learners identifying all three orders successfully.

Part (d)

In part (d), most learners were able to identify that the group was cyclic. Responses that gained fewer marks in part (a) could not gain all marks in part (d). Not all learners gave a suitable reason to support their conclusion. For example, some referred to the orders of the elements all dividing the order of the group; while true, this applies to any finite group and is not sufficient to show that the group is cyclic.

Question 2

Part (a)

This was an accessible question, although it still contained some elements of challenge. In part (a), most learners were able to set up the correct determinant for the characteristic equation, and many completed the part successfully.

Part (b)

Part (b) proved more challenging. Some responses made no progress, either because the part was omitted or because learners did not use the Cayley-Hamilton theorem. In some cases, learners attempted to find $\mathbf{M}^3$ directly and substitute it into the equation; such approaches could not gain credit.

Learners who began by applying the Cayley-Hamilton theorem usually made good progress towards finding the value of a, although some errors occurred in the simplification. Among successful approaches, some learners used the Cayley-Hamilton theorem repeatedly, as intended, to express $\mathbf{M}^3$ in terms of $\mathbf{M}$ before setting up a suitable linear equation. Others found $\mathbf{M}^2$ directly after using the Cayley-Hamilton theorem for the initial stage. Both approaches met the demand of the question and could access all the marks.

Although many learners demonstrated a valid method, notation and layout were often unclear in the working leading to the answer.

Question 3

The use of functions in this question did not appear to cause significant difficulty in parts (a) and (b), which were standard applications of the Euclidean algorithm. However, part (c) proved more challenging, as some learners did not recognise how to apply the algorithm in the general case.

Part (a) and (b)

In parts (a) and (b), many learners performed well. They substituted the given values correctly before applying the Euclidean algorithm and then used back substitution to find the values of x and y. A small number made no progress, while some others omitted the final multiplication by 5 in part (b), despite having completed the preceding work successfully. Incorrect applications of the Euclidean algorithm itself were rare; non-attempts were more common. There were, however, occasional errors in the back substitution.

Part (c)

Part (c) was completed well by some learners, but it was also often omitted by those who were unsure how to approach the general case. Learners who made progress usually completed the first step successfully. However, a few obtained a remainder of -1 and did not explain how this related to the required result. Others obtained the correct remainder but did not comment on its independence of n.

Question 4

This was a challenging question for many learners. However, parts (b) and (c) remained accessible to those who moved on from part (a) and continued with the question. Some learners did not make use of this opportunity, possibly because they did not look ahead to the later parts when they found the proof in part (a) difficult.

The presence of an unknown constant in the closed form may have contributed to this. Some learners appeared to think that the proof in part (a) was intended to find the unknown constant, when in fact this was addressed in part (b), with the result then used in part (c). A more holistic reading of the question would have helped these learners to see how the parts were connected.

Part (a)

The proof by induction in part (a) was made more challenging by the presence of the unknown constant A in the recurrence relation. Many learners allowed this to obscure the standard induction technique. After attempting the base case, many stopped working on the part and moved on. The base case itself also proved problematic for many learners.

Learners who recognised that the presence of the constant did not change the overall induction structure were able to make good progress. These responses often established the base case correctly and made a valid attempt at the inductive step. Some strong, fully successful responses were seen. Induction is often a challenging topic for learners, and the inclusion of the constant appeared to increase the level of demand.

Part (b)

For those who continued with the question, whether they had completed part (a), most were able to answer part (b) successfully and make progress in part (c).

Part (c)

A few responses attempted part (c) by repeated iteration rather than by using the given closed form. These approaches could receive credit where the method was clear.

There were several calculation errors in part (c), including incorrect substitution and arithmetic errors in the final stages. However, many learners were still able to identify the correct final answer.

Question 5

The locus in this question was of a type that learners often find difficult to interpret, and this was reflected in the responses.

Part (a), which asked learners to sketch the locus, was intended to support their approach to the later work. However, this did not always lead to successful progress in part (b). Many learners produced a correct sketch but then made little or no further progress. A few sketched a minor arc rather than a major arc, although some of these learners were still able to make progress in part (b).

Part (a)

Most learners were able to identify the correct locus in part (a), but the quality of the sketches varied. Some diagrams were clear and accurate, while others only just conveyed the intended locus. Overall, learners appeared more confident in identifying the locus than in using it to complete the subsequent work.

Part (b)

Part (b) proved much more challenging. Both methods in the mark scheme were seen, with varying levels of success, although the main method was the more commonly attempted. Some fully correct responses were seen, but many learners only reached the first method mark for finding the distance between the endpoints. This was often part of an otherwise incorrect method. Many learners did not go on to attempt the radius or assumed a radius without showing a valid method.

A common error in the more successful attempts was incorrectly identifying the centre of the circle, often as $(2,4)$ or $(1,2)$. These learners frequently went on to attempt the required distance calculation and could gain method marks. Learners who correctly identified the centre were often able to continue to the correct result.